935

Above formula true for any s $\xrightarrow[\mathcal{L}^{-1}]{}$ transient also possible

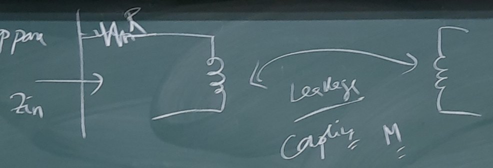
Our goal only at steady state: $s = j\omega$

$$Z_{in} = R_1 + j\omega L_1 - \frac{(j\omega M)(j\omega M)}{R_2 + j\omega L_2 + \boxed{Z_L}} \leftarrow R_L + j\omega X_L$$

$$\left.\begin{array}{l} R_{2L} = R_2 + R_L \\ X_{2L} = \omega L_2 + X_L \end{array}\right\} \quad = R_1 + \underbrace{\frac{\omega^2 M^2 R_{2L}}{(R_{2L}^2 + X_{2L}^2)}}_{\downarrow \text{ resistance increases}} + \underbrace{j\omega L_1 - \frac{j X_{2L}\,\omega^2 M^2}{R_{2L}^2 + X_{2L}^2}}_{\downarrow\ (+)}$$

956

Suppose



Unintended coupling

incr. eff. resistance

Suppose intended coupling:

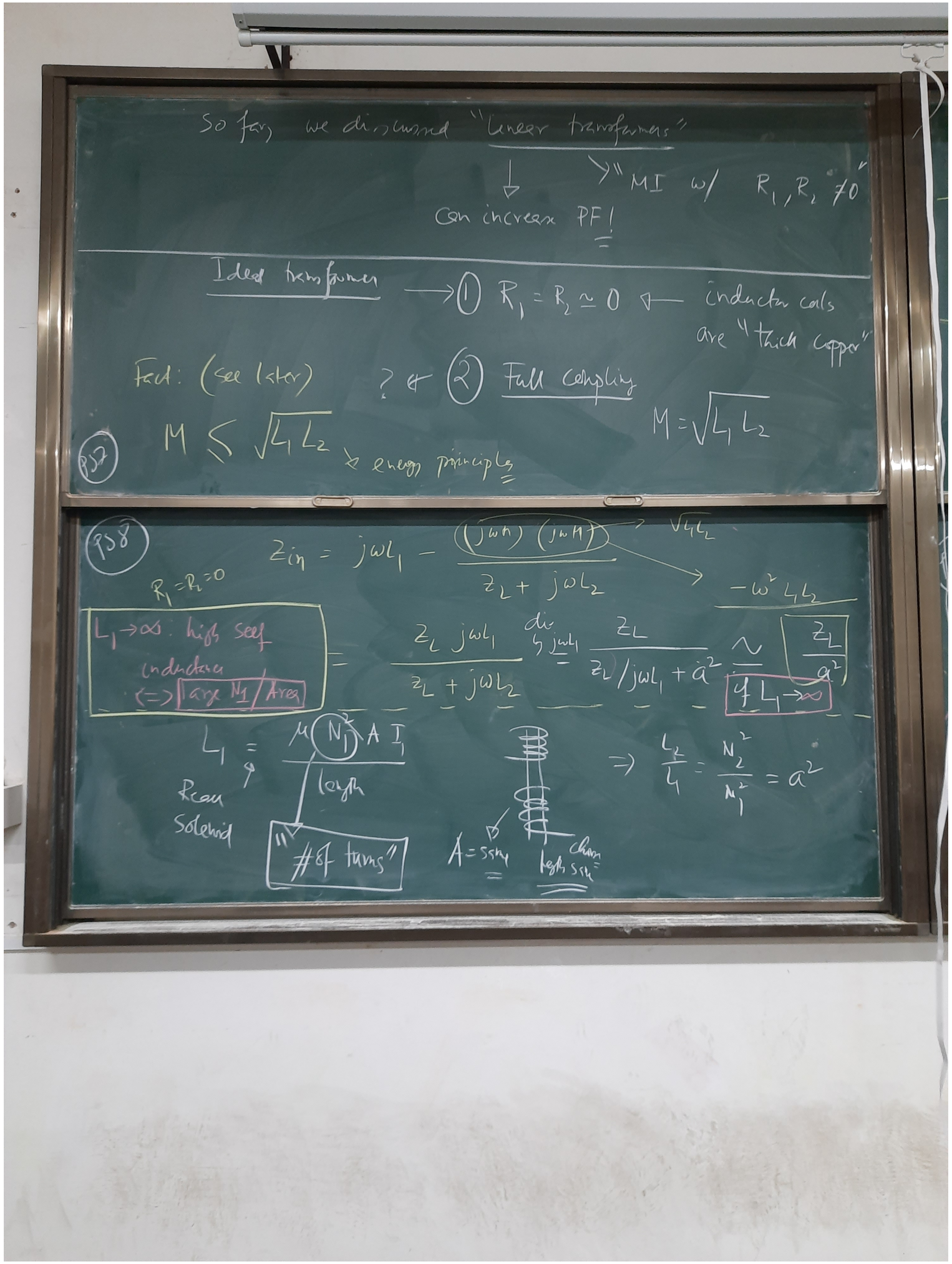
$$PF = \cos\left(\tan^{-1}\frac{\overset{\text{Reactance}}{\boxed{I_n}}}{R_n}\right)$$

Goal is to maximize $\underline{PF}$! (0.k. < true loss is better ↓ last that)

if Z_{in} has minimal "reactance"

$\boxed{\text{Ex. 13.5}}$

Tune "M" ⟶ (+) Small!! ("0" if we can)



So far, we discussed "linear transformers"

↓ can increase PF!

→ "MI w/ $R_1, R_2 \neq 0$"

Ideal transformer → ① $R_1 = R_2 \simeq 0$ ← inductor coils are "thick copper"

Fact: (see later) ? ← ② Full coupling $M = \sqrt{L_1 L_2}$

PS7 $M \leq \sqrt{L_1 L_2}$ ← energy principle

PS8

$$Z_{in} = j\omega L_1 - \frac{(j\omega M)(j\omega M)}{Z_L + j\omega L_2}$$

$\rightarrow \sqrt{L_1 L_2}$

$\rightarrow -\omega^2 L_1 L_2$

$R_1 = R_2 = 0$

$L_1 \to \infty$: high self inductance (⇔) large N_1/Area

$$= \frac{Z_L\, j\omega L_1}{Z_L + j\omega L_2} \overset{\text{div by } j\omega L_1}{=} \frac{Z_L}{Z_L / j\omega L_1 + a^2} \simeq \boxed{\frac{Z_L}{a^2}}$$

if $L_1 \to \infty$

$$L_1 = \frac{\mu N_1^2 A I_1}{\text{length}}$$

Recall solenoid

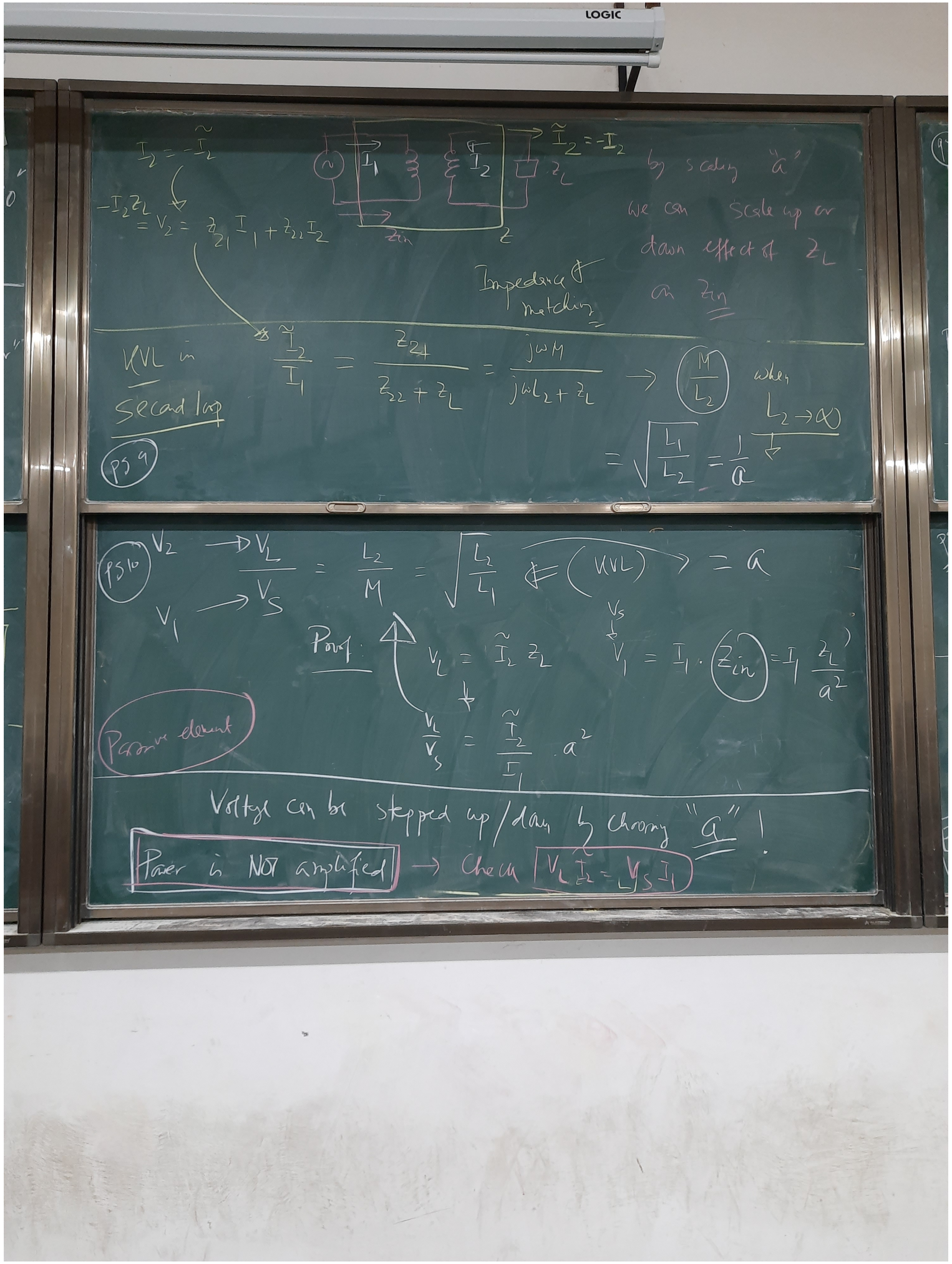
"# of turns"

A = small

length small

chosen

$$\Rightarrow \frac{L_2}{L_1} = \frac{N_2^2}{N_1^2} = a^2$$



$I_2 = -\tilde{I}_2$

$-I_2 Z_L = V_2 = Z_{21} I_1 + Z_{22} I_2$

by scaling "a" we can scale up or down effect of Z_L on Z_{in}

Impedance matching

KVL in second loop

$$\frac{\tilde{I}_2}{I_1} = \frac{Z_{21}}{Z_{22} + Z_L} = \frac{j\omega M}{j\omega L_2 + Z_L} \to \boxed{\frac{M}{L_2}} \text{ when } L_2 \to \infty$$

$$= \sqrt{\frac{L_1}{L_2}} = \frac{1}{a}$$

PS 9

PS 10

$V_2 \to V_L$

$V_1 \to V_S$

$$\frac{V_L}{V_S} = \frac{L_2}{M} = \sqrt{\frac{L_2}{L_1}} \Leftarrow (\text{KVL}) = a$$

Proof:

$V_L = \tilde{I}_2 Z_L$

V_S

$V_1 = I_1 \cdot Z_{in} = I_1 \frac{Z_L}{a^2}$

$$\frac{V_L}{V_S} = \frac{\tilde{I}_2}{I_1} \cdot a^2$$

Passive element

Voltage can be stepped up/down by choosing "a"!

Power is NOT amplified → Check $V_L \tilde{I}_2 = V_S I_1$